

Software Test Plan Document

AI Task Agent Using More Flexible Generic Tools via Model Context Protocol (MCP)

Team Members

Jadon Duff, Computer Science (jduff2024@my.fit.edu)

Jacob Ewasko, Computer Science (jewasko2024@my.fit.edu)

Ioannis Limniatis, Computer Science (ilimniatis2023@my.fit.edu)

Jacob Lebkuecher, Software Engineering (jlebkuecher2023@my.fit.edu)

Faculty Advisor

Eraldo Ribeiro, College of Engineering and Science: Department of Electrical Engineering and
Computer Science (eribeiro@fit.edu)

Client

Doug Armstrong, District Directorate Chief - Response

of

The United States Coast Guard Auxiliary

Table of Contents

1. Introduction
 - 1.1.Purpose
2. Task Execution using Generic MCP Tools Tests
3. Improved AI Knowledge via User-Supplied Documentation Tests
4. Versatile Deployment with Widespread Hardware Compatibility Tests
5. User Interface Tests
6. System Interface Tests
7. Performance Tests
8. Task-Based Performance Tests

1. Introduction

1.1 Purpose

The purpose of this document is to form test cases to test every requirement listed in the Requirements document. The sections of the requirements that are tested are listed above in the Table of Contents.

2. Task Execution using Generic MCP Tools Tests

Implementation:

- Input:

We prompt the agent “Search up on Google Chrome the definition of Earth and return the definition into a text file in my documents folder.”

- Desired Output:

A text file in the user’s documents folder containing the definition of Earth.

If this runs correctly, this satisfies:

ATA-FTE-001	The system shall incorporate an AI Agent via Large Language Model (LLM) that takes textual input and produces textual outputs.
ATA-FTE-002	The system shall expose a Model Context Protocol (MCP) server to the AI Agent, and, via this standardized protocol, allow execution of one of more called tools should the AI Agent produce requests.
ATA-FTE-004	The MCP shall be properly sandboxed, meaning that the MCP tools exposed to the AI Agent do not leave the virtual machine except in specific situations. Specific situations means that the developer intended for that tool to egress from the Virtual Machine and the user provides a form of approval.
ATA-FTE-005	The AI Agent shall map its given task into multiple steps, which shall be logged. The AI Agent shall use “Thinking” or “Reasoning” output modes in its task mapping. These task mappings shall be fully AI generated.
ATA-FTE-006	The MCP server shall be HTTP-based or STDIO-based transport protocol for security.
ATA-FTE-007	The MCP server shall enforce some form of Authentication, which shall be enforced by rotating Bearer token authentication.
ATA-FTE-008	The MCP server shall enforce some form of Authorization, meaning that the tools shall not be executed without prior approval of the User. Note that approval from the User can mean approval during installation and does not require approval for every individual MCP tool.
ATA-FTE-014	The system shall allow the AI Agent to transfer files outside of the virtual machine, such that the user can access these files.

ATA-FTE-017	The system shall allow the AI Agent to output simulated keyboard commands, defined as minimally the keys of a 60% keyboard.
ATA-FTE-026	The system shall allow the AI Agent to get the raw HTML response from a page, if desired by the Agent. Desired by the Agent means the AI Agent requests this in an MCP tool.
ATA-FTE-027	The system shall provide the Accessibility Tree of a webpage, or similar, to the AI Agent. More specifically for “or similar”, means that the webpage will be preprocessed for the AI Agent. Preprocessed means that the webpage raw HTML will be minimally reduced in size or converted to an alternative format that is easier or more efficient for the AI Agent to read.
ATA-FTE-028	The system shall allow the AI Agent to execute REST queries and get a response.
ATA-FTE-029	The system shall allow the AI Agent to execute HTTP queries and get a response.
ATA-FTE-031	The system shall expose the date, the current time, the user’s name, and any other context deemed to be useful by the developers, to the AI Agent, at minimum during the initial prompt.
ATA-FTE-036	The system shall have an egress point of the VM for files, that allow files to be exposed to the user above. The system shall ensure these files cannot be executed by this system on the host machine.
ATA-FTE-039	The system shall allow the AI Agent to complete given Tasks. Given Tasks are user-provided prompts that tell the AI to use the applicable MCP server tools to complete the Task. See the Performance Requirements for the accuracy ratings of Task completion.

Implementation:

- Input:

We prompt the agent “Create a folder called ‘test’, a folder inside of ‘test’ called ‘subtest’ and create a text file inside of ‘subtest’ with the answer to 9+10. Read the contents of the folder ‘test’. Then read the contents of the text file. Update the folder subtest’s name to ‘test1’. Update the text file with the number in that file multiplied by 1000. Move the text file up one directory. Move the folder ‘test1 up one directory. Delete the text file. Delete the folder ‘test1’.”

- Desired Output:

A folder called 'test' is created. A folder called 'subtest' is created inside of 'test'. Then a text file with the path ...\\test\\subtest\\[text file name] is created. The content of this file is the number 19. Then the contents of the folder 'test' are read. Then the contents of the text file are read. Then the folder 'subtest' gets updated to a new name of 'test1'. Then the contents of the text file are updated to 19000. Then the folder arrangement is \\test\\, \\test\\test1, \\test\\[text file name] when moving the file up one directory. Then the folder arrangement is \\test\\, \\test\\[text file name], \\test1\\ when moving the folder up one directory.

If this runs correctly, this satisfies:

ATA-FTE-010	The system shall allow the AI Agent to create, read, update, move, and delete files and folders on its virtual machine.
-------------	---

Implementation:

- Input:

We prompt the agent "Open notepad. Read and summarize what the UI says. Update the contents of a new file 'groceries.txt' to contain '- eggs, milk, bagels' by using notepad and save the file. Move the notepad window to cover exactly the right half of the screen. Then close out of notepad."

- Desired Output:

Notepad opens, the agent gives a summarization of the notepad UI, a new file 'groceries.txt' is created with the contents '- eggs, milk, bagels'. Then the notepad window covers exactly the right side of the screen. Then notepad closes.

If this runs correctly, this satisfies:

ATA-FTE-016	The system shall allow the AI Agent to interact with active windows, defined as the AI Agent can create, read, update, move, and delete active windows on the screen, via simulated keyboard interaction.
-------------	---

Implementation:

- Input:

We prompt the agent "Go to google.com, find the search bar, and search up Wikipedia. Log into my Google account. Go to drive.google.com. Upload a text file containing 'Hello World' to Google Drive. Download that same file from Google Drive."

- Desired Output:

The agent opens a browser and visits google.com. Then from that website, the agent types in the search bar 'Wikipedia'. Then the agent clicks the search button. Then the agent clicks the Sign In

button and logs in with the user's account. Then the agent visits drive.google.com, makes a text file locally containing 'Hello World'. Then the agent uploads the text file. Then the agent downloads the text file.

If this runs correctly, this satisfies:

ATA-FTE-025	The system shall allow the AI Agent to interact with an Internet browser. Interact with an Internet browser means the AI Agent is able to search text queries, ingest pages (URLs), fill out forms, download and upload files, and authenticate when permitted.
-------------	---

Implementation:

- Input:

We prompt the agent "Download python numPy and make a python script that runs on a schedule for every minute for the next 20 minutes, that adds a letter in lexicographic order to a text file each run. Read the contents of the script and copy the contents to the clipboard."

- Desired Output:

The agent should have root access. NumPy is installed and the text file contains "abcdefghijklmnpqrst". Then "abcdefghijklmnpqrst" is copied to the clipboard.

If this runs correctly, this satisfies:

ATA-FTE-003	The MCP Server shall have full control of a virtual machine; with Full Control being defined as sudo (root) access to a common Linux distribution, and execution of any arbitrary text string in the Linux command line.
ATA-FTE-009	The system shall allow the AI Agent to install software on the virtual machine, at a minimum installation via the Command Line of the corresponding virtual machine's operating system.
ATA-FTE-012	The system shall allow the AI Agent to schedule programs to run later, defined as execution at a specific time, assuming the system is running at that time.
ATA-FTE-013	The system shall allow the AI Agent to execute arbitrary programs on its virtual machine, defined as the AI Agent supplying either (1) raw code or (2) the path and name of an executable file.
ATA-FTE-020	The system shall allow the AI Agent to write, read, and execute code, defined in an MCP tool exposed to the AI Agent.

ATA-FTE-022	The system shall allow the AI Agent to add tests to a program, if supported by the programming language the AI Agent is using. The output of these tests shall be automatically routed to the AI Agent. Routed to the AI Agent means the error shall be automatically appended to the next prompt to the AI Agent.
ATA-FTE-024	The system shall allow the AI Agent to install more programming languages if not present.

Implementation:

- Input (Unusual Input):

The user creates a python script called “broken.py” that contains faulty code with the content ‘prin(“hi”)’. Then the user prompts the agent “Run broken.py”

- Desired Output:

Broken.py is run, and an error raises, so the error is queued and routed to AI Task Agent automatically.

If this runs correctly, this satisfies:

ATA-FTE-021	During execution of code, should the program raise an error, this error shall be queued and routed to the AI Agent automatically. Routed to the AI Agent means the error shall be automatically appended to the next prompt to the AI Agent.
-------------	--

Implementation:

- Input:

We prompt the agent “Make a python script that calculates the Fibonacci sequence of 1952 100 times, each time inserting into a 10x10 table into a file named “Fib.txt” inside of the “Fib” folder. While this is happening, monitor running processes and computer performance metrics and place them into an email sent to the same email the user is logged in to. Then read the incoming email, copying the contents to the clipboard. Then read and copy the contents of the Fibonacci table. Then zip the Fibonacci file once complete.”

- Desired Output:

Process and performance metrics email is sent and delivered. Then the contents of the email are copied to the clipboard. Then the contents of the Fibonacci table are copied to the clipboard. The zip file is created in the Fib folder and contains the table when unzipped.

ATA-FTE-011	The system shall allow the AI Agent to monitor running processes on the virtual machine, defined as getting a text output format of the current running processes and ingesting these into the AI Agent's context window.
ATA-FTE-030	The system shall allow the AI Agent to read emails in an email repository and send emails.
ATA-FTE-035	The system shall support ingestion of tabular data, defined as the AI Agent reading a minimum of a 10x10 table of integers.
ATA-FTE-037	The system shall track computer performance metrics and report to the user on demand. Computer performance metrics is defined as the current RAM usage, VRAM usage (if present), and Unified Memory usage (if present).
ATA-FTE-038	The system shall allow the AI Agent to compress folders into ZIP format.
ATA-FAI-007	The system shall automatically manage the context window of the AI Agent. Manage the context window means the system shall not fail when a high context or maximum context is reached, and instead either (1) notify the user and stop the AI Agent or (2) reduce the size of the context window automatically.

Implementation:

- Input:

We prompt the agent "Record the laptop screen for 30 seconds and save the video as "video.mp4". Now watch the video and pause and play the video 5 times throughout the video. Then view the middle of the video and describe what you see and put the description into "description.txt." Then scan the video for any sound and add a description of the sound to "description.txt"

- Desired Output:

The video "video.mp4" should be saved. The agent should pause and play the video five times. The description of the middle part of the video and description of audio should be saved to "description.txt."

ATA-FTE-015	The system shall allow the AI Agent to get a screenshot of the current running operating system (of its virtual machine).
-------------	---

ATA-FTE-018	The system shall allow the AI Agent to ingest imagery or video or video of the current screen (of the virtual machine), defined as instantaneous during tool execution for imagery or instantly to an agent-supplied numerical value of seconds to record video or audio. Instantly or instantaneously is defined as a period of at maximum one second.
ATA-FTE-019	If applications display a system notification, these will be queued and appended to the next prompt to the AI Agent automatically.
ATA-FTE-033	The AI Agent shall support vision inputs, defined as the LLM architecture supporting at minimum JPEG images of 1000x1000 pixel size to be input.
ATA-FTE-034	The AI Agent shall support audio, either through native LLM architecture OR transcription of an audio file.

Implementation:

- Input:

We prompt the agent “Take a screenshot of the current screen and put that image into a pdf file, along with the raw text of that image. Then view the pdf file and print a summary of what the contents of the pdf show into the terminal. Then upload this pdf to git.”

- Desired Output:

There should be a pdf file consisting of an image and the raw text contents of that image. There should be a summary of the pdf file in the terminal output. That pdf should be uploaded to git.

ATA-FTE-023	The system shall allow the AI Agent to interface with Git. Interface with Git means allow the AI to enable Git in a project folder, if not enabled, and execute Git commands.
ATA-FTE-032	The system shall allow the contents of a PDF to be injected into the AI Agent.

3. Improved AI Knowledge via User-Supplied Documentation Tests

Implementation:

- Input:

The user uploads a txt, markdown, pdf, and docx document consisting of the contents of a Wikipedia article on the sun to AI Task Agent. Then the user prompts “How long has the sun existed? Save this information for future use.”

- Desired Output:

The agent returns the answer to the prompt and saves that information for future prompts.

ATA-FAI-001	The system shall integrate Retrieval-Augmented Generation (RAG), defined as an AI model extracting textual content from a user provided text dataset.
ATA-FAI-002	The system shall support TXT, MD, PDF, and DOCX file formats at minimum for RAG, provided by the user.
ATA-FAI-003	The system shall allow the AI Agent to write to a file available for future use, which it can get the contents of via an MCP tool. This allows the AI Agent to have persistent memory outside of its context window as is defined as Dynamic Memory.
ATA-FAI-004	The AI Agent and user shall have the ability to automatically reduce the context window, with reduced context window defined as a minimum of 1 token removed from the context window.
ATA-FAI-007	The system shall automatically manage the context window of the AI Agent. Manage the context window means the system shall not fail when a high context or maximum context is reached, and instead either (1) notify the user and stop the AI Agent or (2) reduce the size of the context window automatically.

Implementation:

- Input:

The user opens a large repository. Then the user prompts “How many instances are there of the word ‘the’?”

- Desired Output:

The agent returns the answer to the prompt by using some form of Fast Context.

ATA-FAI-008	The system shall have some form of Fast Context available to the user and AI Agent, defined as ingestion of textual content to the AI Agent faster than the standard LLM architecture, such as grep and reporting in a project directory instead of ingesting the entire project directory.
-------------	---

Implementation:

- Input (Unusual Input):

The user selects another LLM for AI Task Agent besides the default one.

- Desired Output:

The app displays a disclaimer that says selecting your own LLM comes with risks.

ATA-FAI-009	The system shall allow the user to use another LLM and function the same as using the default LLM provided by the system. Function the same means to meet the requirements in the same manner as before, without penalty should a user select an LLM of lower intelligence. If the user provides an LLM of too low intelligence to complete tasks, or the user provides an LLM of too large size to run efficiently, the system is not required to notify the user of this, but the system must display a disclaimer to the user that selecting their own LLM comes with the risks as mentioned before in this Requirement.
-------------	---

Implementation:

- Input:

The user prompts “How long has the sun existed? Save this information for future use.” Then the user clicks “Clear Memory” button. Then the user prompts “What was the last question I asked you?”

- Desired Output:

The agent returns the answer to the prompt and saves that information for future prompts. Then the user clears the agent’s memory within 2 seconds of the request. Then the agent returns that it does not know what the user is talking about because there is no last question.

ATA-FAI-005	The user shall have the ability to clear the AI memory at any time, with the system complying within two seconds.
-------------	---

Implementation:

- Input:

The user increases the context window of the agent. The user decreases the context window to lower than before the increase. The user uploads a markdown document to AI Task Agent consisting of the contents of a Wikipedia article on the sun.

- Desired Output:

The agent's context window is increased. The agent's context window is decreased. The markdown document is in the agent's dynamic memory.

ATA-FAI-006	The system shall allow the user to manually update the AI Agent's current context window or dynamic memory files.
-------------	---

4. Versatile Deployment with Widespread Hardware Compatibility Tests

Implementation:

- Input:

The user will install AI Task Agent from GitHub to their computer, which is a computer that has adequate hardware requirements, then go through the installation wizard. Then the user installs a local LLM and connects it to AI Task Agent. Then the user runs the prompt “What model are you?”

- Desired Output:

The installation wizard completes successfully and the agent responds to the user prompt and AI Task Agent runs in Normal Operational State.

ATA-FDP-001	The system shall be deployable in a containerized environment, such as Docker.
ATA-FDP-002	The system shall allow the user to define which MCP server tools require manual approval.
ATA-FDP-004	The system shall be downloadable from GitHub.
ATA-FDP-005	The system shall be installable by downloading from GitHub and installing via a wizard.
ATA-FDP-009	The system shall run the LLM locally on the computer system it is installed on.
ATA-FDP-010	The system shall run the MCP server locally on the computer system it is installed on.
ATA-FDP-011	The system shall run the client locally on the computer system it is installed on.
ATA-FDP-013	For systems connected to the Internet, the system can download updates without requiring manual user installation.

ATA-FDP-014	The system shall not interface with any network, such as the Internet, without user approval. Note that user approval can be one-time approval from the user during installation.
ATA-FDP-015	The system shall run in an operational state known as the Normal Operational State more than 98% of the time, when hardware and memory permit the system to run at this state. Normal Operational State means the user can interact with the AI Task Agent as defined in these Software Requirements Specification.

Implementation:

- Input:

The user will install AI Task Agent from GitHub to their computer, then will go through the installation wizard.

- Desired Output:

The installation wizard completes successfully and AI Task Agent successfully downloads.

ATA-FDP-018	The user shall be able to install the system with no action taken other than downloading the files OR transferring via USB AND interacting with the installation wizard.
-------------	--

Implementation:

- Input:

The user will install AI Task Agent from GitHub to their USB, then will go through the installation wizard.

- Desired Output:

The installation wizard completes successfully and AI Task Agent successfully downloads.

ATA-FDP-018	The user shall be able to install the system with no action taken other than downloading the files OR transferring via USB AND interacting with the installation wizard.
-------------	--

Implementation:

- Input:

The user prompts “Return what versions you have installed of Python, Java, C, C++, and an Assembly language.”

- Desired Output:

The agent responds with the versions that are installed of Python, Java, C, C++, and an Assembly language.

ATA-FDP-003	The system shall come preinstalled with Python, Java, C, C++, and an Assembly language.
-------------	---

Implementation:

- Input (Unusual Input):

The user will install AI Task Agent from GitHub to their computer, which is a computer that has inadequate hardware requirements, then go through the installation wizard. Then the user installs a local LLM and connects it to AI Task Agent. Then the user runs the prompt “What model are you?”

- Desired Output:

The installation wizard completes successfully and the agent responds to the user prompt and runs in Degraded Operational State.

ATA-FDP-016	The system shall continue run in the Degraded Operational State. This means that the system is low on memory, running on degraded hardware, or both.
-------------	--

Implementation:

- Input (Unusual Input):

The user will install AI Task Agent from GitHub to their computer, which is a computer that has inadequate hardware requirements, then go through the installation wizard. Then the user installs a local LLM too large to run on their computer and connects it to AI Task Agent. Then the user runs the prompt “What model are you?”

- Desired Output:

The installation wizard completes successfully and the agent responds to the user prompt, and AI Task Agent notifies the user about the performance degradation and advances to Emergency Operational State once memory is maxed out.

ATA-FDP-017	When the system is compromised or the agent is taking undesired actions, an emergency shutdown shall transition the system to the Emergency Operational State within 100 milliseconds. Emergency
-------------	--

	Operational State means that the processes of the AI Task Agent shall be terminated. The system shall transition to this state upon user input.
ATA-FDP-020	The system shall stop and notify the user if it is detecting a too large context window or other performance degradation.

Implementation:

- Input:

The user will install AI Task Agent from GitHub to a USB to be used for offline purposes, then go through the installation wizard, disabling browser tools. Then the user installs a local LLM and connects it to AI Task Agent. Then the user runs the prompt “What model are you?”

- Desired Output:

The installation wizard completes successfully onto the USB. The agent responds to the user prompt.

ATA-FDP-006	The system shall be installable by putting its installation wizard onto USB.
ATA-FDP-007	The system shall be installable on a computer with no Internet access.
ATA-FDP-008	The system shall allow the user to remove tools from the MCP server, hiding them from the AI Agent. This is done during installation of the system.
ATA-FDP-012	The system shall run and meet all applicable requirements, when installed on a computer system that does not have Internet access. Applicable requirements means all requirements that do not require Internet access.
ATA-FDP-019	The system shall contain all applicable files if installing via USB, such as the LLM weights.

5. User Interface Tests

Implementation:

- Input:

The user uploads documentation on how to work with csv files, inputs a csv file with 100 random numbers into the prompt, and prompts “What is the average value of the numbers in this file?”

- Desired Output:

The agent returns the average value of the numbers.

ATA-IUI-001	The system shall allow a user to send textual data to the AI Agent through the GUI.
ATA-IUI-003	The user shall interact with the AI Agent via a Graphical User Interface (GUI) and shall submit tasks with a prompt input button.
ATA-IUI-004	The system shall accept media such as files, images, and video from the User during prompting for task completion.
ATA-IUI-005	The system shall provide a user interface for uploading documentation for the RAG subsystem.

Implementation:

- Input:

The user explicitly allows AI Task Agent to log analytics or metrics data. Then the user prompts “Make a python script that returns ‘Hello World’.”

- Desired Output:

The agent should make the python script and run it. The system should log analytics or metrics data.

ATA-IUI-002	The system shall not log analytics or metrics data without explicit user approval. Note that approval from the User can mean approval during installation and does not require approval for every analytics or metrics log.
-------------	---

Implementation:

- Input:

The user explicitly does not allow AI Task Agent to log analytics or metrics data. Then the user prompts “Make a python script that returns ‘Hello World’.”

- Desired Output:

The agent should make the python script and run it. The system should not log analytics or metrics data.

ATA-IUI-002	The system shall not log analytics or metrics data without explicit user approval. Note that approval from the User can mean approval during installation and does not require approval for every analytics or metrics log.
-------------	---

Implementation:

- Input:

The user uploads an image of a spreadsheet, and prompts “What is the average value of the numbers in this file?”

- Desired Output:

The agent returns the average value of the numbers.

ATA-IUI-004	The system shall accept media such as files, images, and video from the User during prompting for task completion.
-------------	--

Implementation:

- Input:

The user uploads a video of multiple spreadsheets, and prompts “What is the value in these files?”

- Desired Output:

The agent returns the most common value in these files.

ATA-IUI-004	The system shall accept media such as files, images, and video from the User during prompting for task completion.
-------------	--

Implementation:

- Input (Unusual Input):

The user will install AI Task Agent from GitHub to their computer, which is a computer that has inadequate hardware requirements, then go through the installation wizard. Then the user installs a local LLM too large to run on their computer and connects it to AI Task Agent. Then the user runs the prompt “What model are you?” Then the user interacts with the UI to enable emergency shutoff.

- Desired Output:

The installation wizard completes successfully and the agent responds to the user prompt, and AI Task Agent notifies the user about the performance degradation, and the emergency shutdown phase is enabled by user input.

ATA-IUI-006	A User Interface shall be provided for other requirements regarding user interaction, such as emergency shutoff.
ATA-IUI-007	The system shall expose model hyperparameters such as temperature to the user, and be selectable via a GUI.
ATA-IUI-011	The system shall expose an installation wizard to the user for installation of the software system.
ATA-IUI-016	The system shall allow the user to upload a custom GGUF file of an LLM. Allow means provide a means of uploading or pointing to a file location of the GGUF.

Implementation:

- Input:

The user deletes AI Task Agent by clicking a single button with confirmation.

- Desired Output:

AI Task Agent and its files are deleted.

ATA-IUI-012	The user shall be able to delete the software completely from the user's computer via a single UI element and a confirmation.
-------------	---

Implementation:

- Input:

The user prompts "Write me an essay about the sun". Then the user clicks their way through the UI to try and get past chats and running tasks of that chat. Then the user clicks their way back to their active chat and running task.

- Desired Output:

The agent starts answering the prompt. Then the UI brings the user to the desired past chat and displays the running tasks of that chat. Then the UI brings the user to the active chat and displays the running tasks of that chat.

ATA-IUI-008	The GUI shall display past and active running tasks from the AI Agent.
ATA-IUI-009	The GUI shall display past and active prompt/chat windows with the AI Agent.

Implementation:

- Input (Unusual Input):

The user will install AI Task Agent from GitHub to their computer, then go through the installation wizard. Then the user installs a local LLM and connects it to AI Task Agent. Then the user runs the prompt “What model are you?”. Then the user changes the prompt to “What does google.com show?”. Then the user changes to air-gapped mode and tries the same prompt again “What does google.com show?”

- Desired Output:

The installation wizard completes successfully and the agent responds to the first user prompt, then responds to the second user prompt. Then the agent returns an error to the user that it cannot access the browser because it is in air-gapped mode.

ATA-IUI-010	The GUI shall allow the user to perform creation, reading, updating, deletion, and exporting of AI Agent task interactions (e.g. prompting).
ATA-IUI-014	The system shall require a single UI interaction to designate an air-gapped system, disabling Internet and other networked/communication abilities.
ATA-IUI-015	The system shall have a conversion UI element exposed to the user that allows the system to be converted to an air-gapped system. The system shall NOT allow conversion from air-gapped to networked and instead would require reinstallation of the system.

Implementation:

- Input:

The user prompts “What model are you?”. Then the user exports the prompt and answer to the prompt. Then the user deletes their prompt.

- Desired Output:

The agent should respond to the prompt. Then the system should export the data to a place of the user’s choice on their device. Then the system should delete the prompt and the prompt response.

ATA-IUI-010	The GUI shall allow the user to perform creation, reading, updating, deletion, and exporting of AI Agent task interactions (e.g. prompting).
-------------	--

Implementation:

- Input:

The user prompts “How much memory do I have?” Then the user turns off performance tracking approval. Then the user prompts “How much memory do I have?”

- Desired Output:

The agent responds to the first user prompt but returns an error to the user stating that it does not have access, then requests approval.

ATA-IUI-013	The user shall have the ability to revoke given approvals at any time, unless the tasks requiring approvals have already occurred for AI Agent tasks.
-------------	---

Implementation:

- Input (Unusual Input):

The user will install AI Task Agent from GitHub to their computer, then go through the installation wizard, initialized to air-gapped mode. Then the user attempts to change to networked mode.

- Desired Output:

The installation wizard completes successfully in air-gapped mode. The system then prompts the user saying air-gapped to networked is not allowed and would require reinstallation of the system.

ATA-IUI-015	The system shall have a conversion UI element exposed to the user that allows the system to be converted to an air-gapped system. The system shall NOT allow conversion from air-gapped to networked and instead would require reinstallation of the system.
-------------	--

6. System Interface Tests

Implementation:

- Input:

The user prompts “Check how many repositories are on my GitHub account and store the result in my documents folder.”

- Desired Output:

The agent finds the result of the number of GitHub repositories on the user's account and places it in the user's documents folder.

ATA-ISI-001	The system shall use the host's Internet connection to transmit and receive data to other networked systems if desired by the user. Details of what is transmitted/received via the Internet is detailed in the functional requirements.
ATA-ISI-004	The system shall provide a single ingress and egress point with the host and virtual machine. This single ingress and egress point shall be a directory shared by both the host machine and the virtual machine.

Implementation:

- Input:

The user allows the system to query GitHub for automatic AI Task Agent updates. A new AI Task Agent update comes out.

- Desired Output:

The system queries GitHub releases API for a newer designated release of the AI Task Agent system. The user is notified and the update automatically downloads.

ATA-ISI-002	The system shall query GitHub upon user approval for automatic updates. Query means check the GitHub releases API for a newer designated release of the AI Task Agent system. If so, the user shall be notified and if approved shall download the new update. This requirement is only for Internet-connected versions of the system.
-------------	--

Implementation:

- Input:

The user will install AI Task Agent from GitHub to their computer, then go through the installation wizard. Then the user installs a local LLM and connects it to AI Task Agent. Then the user runs the prompt "What are my important emails for today? Also install python NumPy".

- Desired Output:

The installation wizard completes successfully. LLM weights are downloaded after the user connects their LLM. The agent responds with emails that are deemed important within the last 24 hours. Then the agent installs NumPy and pins NumPy's library version.

ATA-ISI-003	The system shall integrate with an email or other messaging service. Integrate with means allow the AI to use at a minimum the minimum
-------------	--

	required features of such messaging service to transmit and receive messages.
ATA-ISI-005	The system shall download applicable external subsystems if they are too large to commit to the Git repo of this project, during installation. For example, the LLM weights would be too large to upload to GitHub, so during installation the system would download the file. This is not required during USB installation, as the files would be located on the USB.
ATA-ISI-006	The system shall pin library versions if installing via an external package provider, such as PIP.

Implementation:

- Input:

The user prompts “Send an email from the user account to the same user account.”

- Desired Output:

The agent sends the email and receives the email.

ATA-ISI-003	The system shall integrate with an email or other messaging service. Integrate with means allow the AI to use at a minimum the minimum required features of such messaging service to transmit and receive messages.
-------------	--

7. Performance Tests

Implementation:

- Input:

The user will install AI Task Agent from GitHub to their MacOS computer which has the hardware specifications of 18-core CPU, 20-core GPU, 307 GB/s memory bandwidth, 64 GB unified memory or RAM+VRAM equivalent, 1 TB storage, 100-watt-hour battery, visual display, and USB port, then go through the installation wizard. Then the user runs the prompt “What are my important emails for today? Also install python NumPy”.

- Desired Output:

The installation wizard completes successfully (max size of 80GB and 10-40 billion parameter LLM, 10 minutes initialized, 90 minutes full) and the agent responds (within 15 seconds) with emails that are deemed important within the last 24 hours. The agent produces at least an average 7 tokens per second. Then the agent installs NumPy. The agent finishes in under 20 minutes.

ATA-PFR-001	The system shall meet all requirements on a hardware machine consisting of a minimum of 18-core CPU, 20-core GPU, 307 GB/s memory bandwidth, 64 GB unified memory or RAM+VRAM equivalent, 1 TB storage, 100-watt-hour battery, visual display, and USB port. The system may work on lower hardware specifications but must meet requirements on these specifications. These specifications mean the system is intended to work on a medium-to-high-end consumer hardware.
ATA-PFR-002	The system shall function on a host machine running MacOS. Function means it meets all requirements.
ATA-PFR-004	The system shall be of a maximum size of 80 GB, including the LLM weights.
ATA-PFR-005	The system shall include an LLM of 10-40 billion parameters.
ATA-PFR-006	The AI Agent shall produce tokens at a minimum average rate of 7 tokens per second, per each of the Task examples in the Task-Based Performance Requirements.
ATA-PFR-007	The AI Agent shall complete each Task-Based Performance Requirements within a timeframe of 20 minutes. Meaning, 20 minutes per requirement in the Task-Based Performance Requirements.
ATA-PFR-008	A User shall download the system from the Internet in a maximum of 10 minutes for initializing the download and 90 minutes for full download (including LLM weights). This requirement may be attempted multiple times if external systems hosting dependencies such as the LLM weights are down/slow.
ATA-PFR-010	A User shall submit a Task to the AI Agent within 15 seconds, with the AI Agent starting on the Task within these 15 seconds, not counting the time to type in the prompt.

Implementation:

- Input:

The user closes AI Task Agent. The user starts up AI Task Agent.

- Desired Output:

AI Task agent closes. AI Task Agent starts up within 120 seconds.

ATA-PFR-009	The software shall start after being closed within a period of 120 seconds, not including the time to load the LLM weights into memory.
-------------	---

Implementation:

- Input:

A New User will install AI Task Agent from GitHub to their Windows computer which has the hardware specifications of 18-core CPU, 20-core GPU, 307 GB/s memory bandwidth, 64 GB unified memory or RAM+VRAM equivalent, 1 TB storage, 100-watt-hour battery, visual display, and USB port, then go through the installation wizard. Then the user installs a local LLM (max size of 80GB and 10-40 billion parameters) and connects it to AI Task Agent. Then the user runs the prompt (within 5 minutes) a prompt of their choice.

- Desired Output:

The installation wizard completes successfully and the agent responds with emails that are deemed important within the last 24 hours.

ATA-PFR-003	The system shall function on a host machine running Windows. Function means it meets all requirements.
ATA-PFR-011	A New User shall install the system without consulting the developers, faculty advisor, or client. A New User is defined as a user who has never used the system, or watched the system being used by another user for more than 60 seconds.
ATA-PFR-012	A New User shall open the application and submit a Task to the AI Agent within 5 minutes. A New User is defined as a user who has never used the system, or watched the system being used by another user for more than 60 seconds.

Implementation:

- Input:

The user installs AI Task Agent from GitHub onto their USB and goes through the installation wizard.

- Desired Output:

Then AI Task Agent is installed on the USB drive within 10 minutes.

ATA-PFR-013	The system shall transfer and install via USB within 10 minutes.
-------------	--

8. Task-Based Performance Tests

Implementation:

- Input:

The user will install AI Task Agent from GitHub to their computer, then go through the installation wizard. Then the user installs a local LLM and connects it to AI Task Agent. Then the user runs the prompt “What are my important emails for today? Also download Python NumPy on my USB drive. Then write a python script that prints “Hello World” in the terminal. Then run that script. Then copy that output. Then open Notepad. Then paste the output from the terminal output into a text file on Notepad. Then delete ‘World’ in the Notepad file. Then paste the output again”.

- Desired Output:

The installation wizard completes successfully and the agent responds with emails that are deemed important within the last 24 hours. Then the agent installs NumPy on the USB drive. Then the agent creates a Python script, runs it, then copies the output. Then the agent opens Notepad and pastes the terminal output “Hello World.” Then the agent deletes ‘World’ and the file is left with “Hello ”. Then the agent pastes the output again and is left with the contents of “Hello Hello World”.

ATA-OPS-001	The AI Agent shall reason during each Task and identify any subtasks it needs to complete.
ATA-OPS-002	The AI Agent shall complete Tasks requiring a minimum of seven subtasks or steps that must be completed in sequential order to complete.
ATA-OPS-005	The AI Agent shall complete Tasks requiring a minimum of three steps and performing create, update, read and delete operations on files.
ATA-OPS-006	The AI Agent shall complete Tasks requiring a minimum of three steps and interacting with an open application, such as Notepad.
ATA-OPS-007	The AI Agent shall complete Tasks requiring a minimum of three steps and moving the mouse and pressing the simulated keyboard.
ATA-OPS-011	The AI Agent shall complete Tasks requiring a minimum of three steps and create a small programming project from scratch.

ATA-OPS-013	The AI Agent shall complete Tasks requiring a minimum of three steps and search up information not present in its knowledge base.
-------------	---

Implementation:

- Input:

The user runs the prompt “Run three terminal commands in this order: pwd, cd .., ls. Then install Python NumPy, Python Matplotlib, and Python PyTorch. Then make a python script that prints Hello World and run it three times.”.

- Desired Output:

The agent prints the working directory, changes directory up a level, and lists the contents of the directory. Then the agent installs Python NumPy, Python Matplotlib, and Python PyTorch. Then the Hello World Python script is made and is ran three times.

ATA-OPS-003	The AI Agent shall complete Tasks requiring a minimum of three steps and running terminal commands.
ATA-OPS-004	The AI Agent shall complete Tasks requiring a minimum of three steps and installing software.
ATA-OPS-008	The AI Agent shall complete Tasks requiring a minimum of three steps and write and execute code.
ATA-OPS-014	The AI Agent shall complete Tasks requiring a minimum of three steps and state a fact not found in its knowledge base, instead in the provided RAG context.

Implementation:

- Input:

The user runs the prompt “Go to YouTube, click on any video, describe what you see, and create a new YouTube account with an email that you create.”.

- Desired Output:

The agent prints the working directory, changes directory up a level, and lists the contents of the directory. Then the agent installs Python NumPy, Python Matplotlib, and Python PyTorch. Then the Hello World Python script is made and is ran three times.

ATA-OPS-009	The AI Agent shall complete Tasks requiring a minimum of three steps and interact with imagery/video.
-------------	---

ATA-OPS-012	The AI Agent shall complete Tasks requiring a minimum of three steps and navigate to a page on a browser.
ATA-OPS-015	The AI Agent shall complete Tasks requiring a minimum of three steps and fill out an online form.
ATA-OPS-016	The AI Agent shall complete Tasks requiring a minimum of three steps and submit data to a website after logging in.

Implementation:

- Input:

The user uploads faulty code to the GitHub repository 'agenttest', then runs the prompt "I have an error in the code in the GitHub repository 'agenttest'. Please fix it".

- Desired Output:

The agent finds the repository 'agenttest', finds the code issue and fixes the code issue.

ATA-OPS-010	The AI Agent shall complete Tasks requiring a minimum of three steps and fix a small bug in a local repository.
-------------	---

Implementation:

- Input:

The user runs the prompt "Order pencils from amazon and send it to my address".

- Desired Output:

The agent finds the pencils on Amazon, buys them and ships them to the user's address.

ATA-OPS-017	The AI Agent shall complete Tasks requiring a minimum of three steps and interacting with an Internet based system that changes something in the physical world. For example, ordering a package from an online shopping center.
-------------	--

Implementation:

- Input:

The user runs the prompt "Email the developers, faculty advisor, professor, and client of the Florida Tech Computer Science Senior Design AI Task Agent project and say you completed requirement 'ATA-OPS-018'".

- Desired Output:

The agent finds the emails and sends the email.

ATA-OPS-018	The AI Agent shall email its developers, the faculty advisor, the professor, and the client a summary that it has completed this requirement.
-------------	---

Implementation:

- Input:

The user runs the prompt “How big is the Sun? Remember this for future tasks”. Then the user clears AI Task Agent’s memory. Then the user prompts “How big is the Sun? Remember this for future tasks”. Then the user prompts “Summarize the last question I asked you.”

- Desired Output:

The agent saves the fact for memory, then memory is cleared, then the fact for memory is saved again, then the last question is remembered and is summarized.

ATA-OPS-019	The AI Agent shall be given a task to remember facts for future chats, intending for the AI Agent to save the given fact in its writable memory. The AI Agent’s memory shall then be cleared. The AI Agent shall then be given another Task and is expected to reason to pull from its writable memory (dynamic memory).
-------------	--

Implementation:

- Input (Unusual Input):

The user copies and pastes 1000 characters of Lorem Ipsum 1000 times into the prompt and runs.

- Desired Output:

The agent interprets the user’s message.

ATA-OPS-020	The AI Agent shall be given a prompt that exceeds its context window, and the system shall not crash.
-------------	---

Implementation:

- Input:

The user runs the prompt “Make an X account and make a post saying ‘Hello World’.”

- Desired Output:

The agent makes an X account using the user’s email and makes the post.

ATA-OPS-021	The AI Agent shall complete Tasks requiring a minimum of three steps and modifying an Internet artifact that is viewable to the entire Internet.
-------------	--

Implementation:

- Input:

The user runs the prompt “Complete 3 CTF challenges on a reputable cybersecurity website.”

- Desired Output:

The agent completes 3 challenges on a reputable cybersecurity website.

ATA-OPS-022	The AI Agent shall complete Tasks requiring a minimum of three steps and complete a Capture the Flag (CTF) cybersecurity challenge. The CTF shall be found from a reputable cybersecurity website.
-------------	--

Implementation:

- Input:

The user runs the prompt “Do this three times: (Complete 3 CTF challenges on a reputable cybersecurity website. Then make an X account and make a post saying ‘Hello World’. Then run three terminal commands in this order: pwd, cd .., ls. Then install Python NumPy, Python Matplotlib, and Python PyTorch. Then make a python script that prints Hello World and run it three times.) ”

- Desired Output:

The agent does the three tasks three unique times.

ATA-OPS-023	The AI Agent shall complete this requirement 3 times with unique Task-Based Performance Requirements each attempt. To complete, 2 of the other Task-Based Performance Requirements shall be chosen, and the AI will be required to complete both in the same prompt.
-------------	--

Implementation:

- Input:

A BQ or AUXOP user runs 2 prompts of their choosing or prompts “What do I do when I see an abandoned ship?” and “How long can I expect a human to tread water?”

- Desired Output:

The agent does what the user wants to do or describes what to do when seeing an abandoned ship and describes how long a human can tread water.

ATA-OPS-024	The AI Agent shall take a question from a person with a status of Basically Qualified (BQ) or Operational Auxiliarist (AUXOP) in the Coast Guard Auxiliary, and output a correct answer, given Coast Guard Auxiliary documentation.
ATA-OPS-026	The AI Agent shall perform 2 Tasks accurately in an operational environment for the Coast Guard, or nonoperational environment reporting official hours to the US Coast Guard Auxiliary. If an operational environment is unavailable OR a nonoperational environment in which the User is reporting official hours via the 7029 form, the AI Agent shall instead perform 10 Tasks accurately in a simulated operational environment. Operational environment means an environment in which the User is under official government orders, is on a boat, is in an aircraft, is performing an official Coast Guard mission. Nonoperational environment reporting official hours means the User is in a state in which they are logging hours that are reported on the US CGAUX ANSC 7029 Form.

Implementation:

- Input:

User inputs search and rescue plan document which is known to be “invalid” (taking the user through a thunderstorm or other weather concern) and asks the ai agent to advise on the SAR plan based on the weather given coordinates.

- Desired Output:

The agent successfully advises against flying through the bad weather by searching the weather using the MCP tools and rerouting to a non-inclement weather zone. 4 tasks: searching up content, inputting coordinates/location, rerouting user, explaining SAR advised plan.

ATA-OPS-025	The AI Agent shall complete a minimum of 4 Tasks correctly given by the Client, who is part of the Coast Guard Auxiliary.
-------------	---

Implementation:

- Input:

The Faculty Advisor user runs the prompt of their choosing containing a minimum of 4 tasks.

- Desired Output:

The agent accomplishes the desired tasks.

ATA-OPS-027	The AI Agent shall complete a minimum of 4 Tasks correctly given by the Faculty Advisor.
-------------	--

Implementation:

- Input:

The Florida Institute of Technology CSE4101 CS Projects 1 Professor user runs the prompt of their choosing containing a minimum of 4 tasks.

- Desired Output:

The agent accomplishes the desired tasks.

ATA-OPS-028	The AI Agent shall complete a minimum of 4 Tasks correctly given by the Florida Institute of Technology CSE4101 CS Projects 1. If the professor is unavailable, another professor may give the 4 Tasks.
-------------	---